

DOCIPHER

AI-Powered Medical Intelligence

ABOUT US

OUR CONTACT

AI ASSISTANT

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DOCIPHER

PROBLEM STATEMENT

Complex Data

Medical reports are complex, technical, and difficult for patients to understand.

Fragmented History

Patient histories are fragmented, making them hard to manage.

Impersonal Answers:

Getting simple, personalized health answers is difficult.

THE SOLUTION: "DOCIPHER"

- Summarizes reports with AI.
- Organizes a central patient profile.
- Answers questions with a personalized AI chatbot.
- Manages medication reminders.



KEY FEATURES

01. AI Report Summarizer

Upload medical reports (Text, PDF, JPG, PNG) and receive simple summaries.

02. Personalized AI Chatbot:

An assistant (powered by Google Gemini) that answers questions using the context of your personal health profile.

03. Comprehensive Patient Profile

A 20+ field form to store everything: vitals, allergies, chronic conditions, medications, and emergency contacts.

04. Medication Reminders

A full CRUD (Create, Read, Update, Delete) system for scheduling medication.

05. PDF & Data Export

Download your complete profile or any AI-generated summary as a PDF.

06. Comprehensive Patient Profile

A 20+ field form to store everything: vitals, allergies, chronic conditions, medications, and emergency contacts.

TECHNOLOGY & ARCHITECTURE

Technology Stack:

Backend: Python, Django

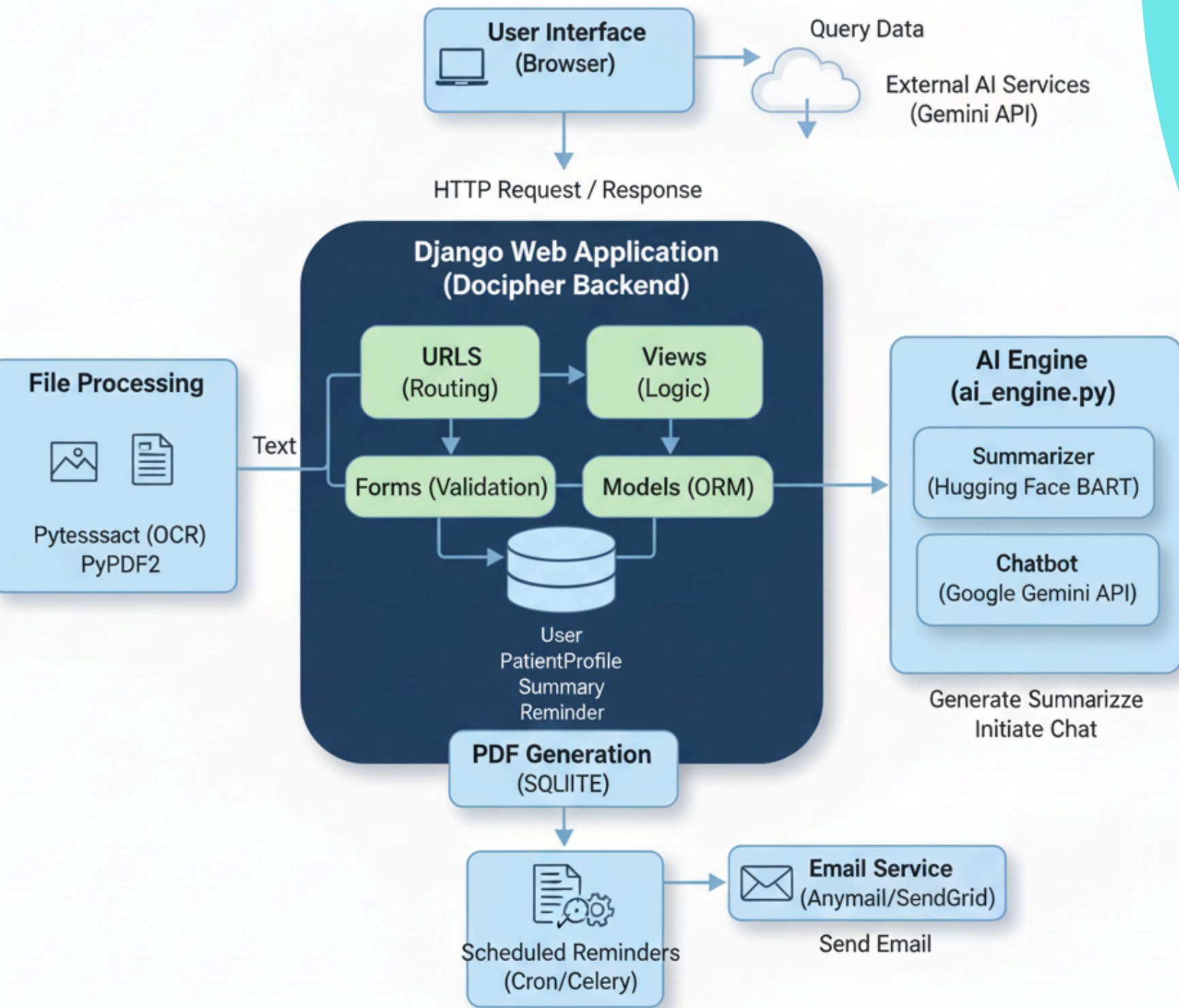
AI/ML: Google Gemini, Hugging Face Transformers, Pytesseract (OCR)

Frontend: HTML, Bootstrap 5, JavaScript (AJAX), CSS

DB & Utils: SQLite, xhtml2pdf

Project Structure (summarizer app):

- **models.py:** Database Structure (The "Brain")
- **views.py:** Business Logic (The "Actions")
- **ai_engine.py:** AI Logic (The "Intelligence")
- **forms.py:** Data Validation (The "Rules")
- **templates/:** HTML Files (The "Face")



WORKFLOW & ADVANCED FEATURES

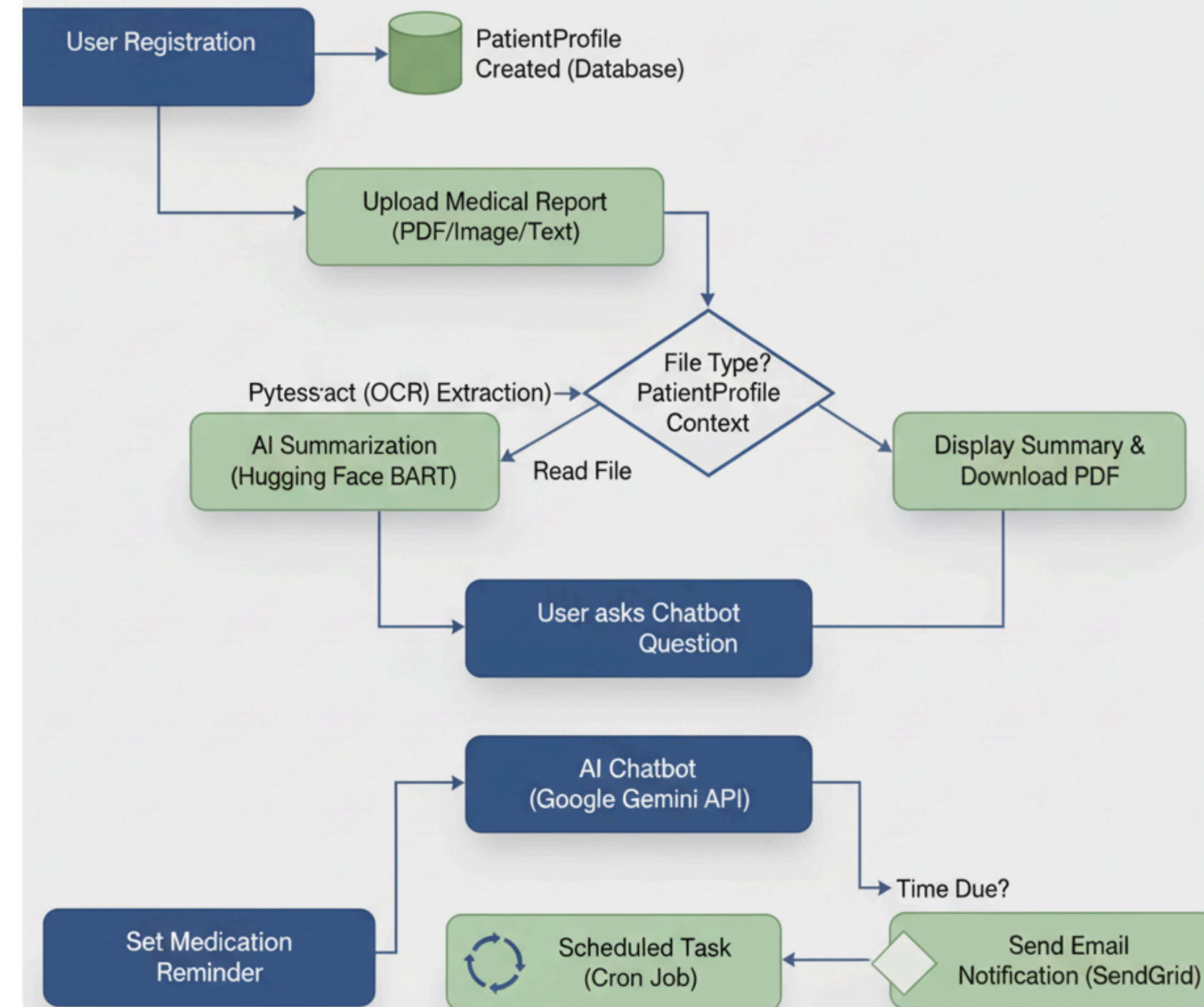
Example User Workflow

1. User registers (Profile is auto-created).
2. User uploads a PDF report -> Gets an AI summary.
3. User asks chatbot a question -> Gets a personalized answer based on their profile.
4. User sets a medication reminder.

Advanced Features

- PDF Generation: xhtml2pdf converts HTML templates (like profile_pdf.html) into downloadable PDFs.
- Custom Admin Panel: Uses django-jazzmin with custom filters and search for easy management.
- Automated Reminders: A custom command (send_reminders.py) can be run on a schedule (CRON job) to send email alerts.

Docipher User Workflow



OUR SECURITY & FUTURE SCOPE

SECURITY



@LOGIN_REQUIRED

decorator on all sensitive views.



DATA ISOLATION

All database queries are filtered by
`user=request.user`



API KEYS

stored securely in .env file



CSRF PROTECTION

and built-in SQL Injection prevention

FUTURE SCOPE



DEPLOY TO PRODUCTION
(E.G., AWS/HEROKU) WITH A
POSTGRESQL DATABASE.



USE CELERY & REDIS FOR
ASYNCHRONOUS
(BACKGROUND) REMINDERS.



IMPROVE OCR FOR
HANDWRITTEN NOTES.

THANK YOU AND STAY HEALTHY

QUESTION ?